RAJAT PATEL

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SUMMARY

Senior ML Engineer with 6+ years shipping production NLP systems (NER, entity resolution, information extraction, knowledge graph enrichment, LLM-powered automation). Published researcher (EMNLP, RANLP) with a track record of turning messy real-world unstructured data into structured signals and delivering human-in-the-loop ML pipelines from prototype to production.

SKILLS

Languages: Python, Go, JavaScript/TypeScript, SQL, Shell Scripting
ML & Frameworks: TensorFlow, PyTorch, FastAPI, Flask, Django, AWS SageMaker, Kubeflow, MLflow, Ray
Data & Streaming: Apache Flink, AWS Glue, Kafka, Prefect, Snowflake
Tools & Databases: Milvus, DynamoDB, Postgres, MongoDB, Kubernetes, Docker, Arize, DataDog

EXPERIENCE

Senior Machine Learning Engineer | *Chime Financial Inc.* Nov 2023 - Present

- Built a scalable LLM-powered labeling automation system combining auto-labeling (LangGraph) with human-in-the-loop review for financial data: 150% improvement in throughput, turnaround reduced from weeks to hours, 90% reduction in manual review load, 99% reduction in annotation cost, and 4+ week acceleration in model iteration cycles (SageMaker, Snowflake, Datadog, Hex).
- Owned end-to-end ML model improvements across 12M+ members (data sourcing, sampling, training, experimentation, and rollout), improving precision/recall and driving meaningful business impact across product and operations.
- Prototyped sequential transformer and graph-based (GNN) approaches for structured signal extraction from noisy, high-dimensional real-world data, exploring foundation-model-inspired modeling beyond traditional rule and feature pipelines.
- Improved ML infrastructure across training and production: integrated MLflow for experiment traceability, built near-real-time data pipelines (Apache Flink, AWS Glue, DynamoDB) at $<$30ms latency, and used Arize for model observability; partnered with ML Platform to shape the ML Infrastructure roadmap.
- Acted as technical lead and team mentor: coached ICs and junior engineers through major model launches, led data-driven root-cause investigations, and provided cross-functional guidance to Product and Engineering partners.

Senior Machine Learning Engineer | *Interos Inc.* Jan 2020 - Nov 2023

- Developed and productionized Transformer-based NER + event extraction pipelines to extract structured signals from noisy unstructured text, and built human-in-the-loop feedback/retraining loops (SageMaker GroundTruth) to handle concept drift and continuously enrich the Interos knowledge graph.
- Shipped a deep-learning entity dedup/linking pipeline (contrastive candidate generation, cross-encoder reranking, Milvus vector search) achieving $\sim$90% top-10 recall, enabling reliable entity resolution across millions of noisy real-world records.
- Co-led deployment and operation of large-scale Milvus vector search over $\sim$400M entities, tuning indexing and HNSW-based retrieval to meet latency/availability requirements across multiple ML products.
- Built and owned a full-stack active learning pipeline for large-scale visual classification (web scraping, active learning, ImageNet-based search), reducing manual identification time by $>$80% through iterative human-in-the-loop annotation and model retraining.

EDUCATION

University of Maryland Baltimore County May 2020
Master of Science in Computer Science
Shivaji University May 2014
Bachelors of Engineering in Electronics and Telecomm. Engg.

PUBLICATIONS

- On the Complementary Nature of Knowledge Graph Embeddings, Fine Grain Entity Types and Language Modeling. InProceedings, EMNLP Workshop on Deep Learning Inside Out, November 2020
- InterosML @ Causal News Corpus 2023: Understanding Causal Relationships: Supervised Contrastive Learning for Event Classification. In Proceedings, RANLP CASE 2023 Workshop, September 2023